Poisson Distribution

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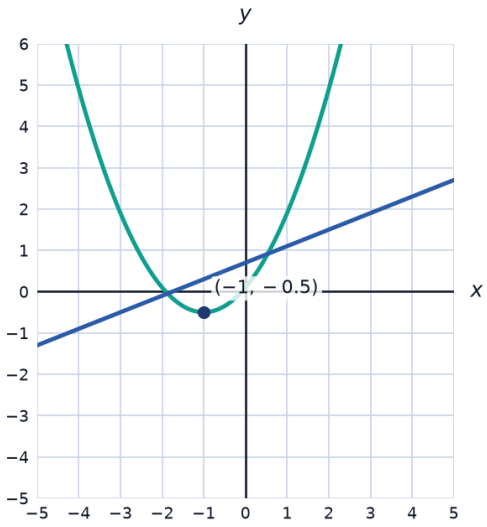
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Conditions for a Poisson distribution**

State the key conditions required for a random variable to be modeled by a Poisson distribution.

2 **[Foundational]** • **Target: Conditions for a Poisson distribution**

Explain why the number of texts a person receives in a randomly chosen hour might reasonably be modeled by a Poisson distribution, but the number of texts received right around a pre-arranged deadline might not.



3 **[Foundational]** • **Target: Computing Poisson probabilities**

A shop receives an average of 3 customers per minute, modeled by $X \sim \text{Po}(3)$. Find $P(X = 5)$, to three decimal places.

4 **[Foundational]** • **Target: Computing Poisson probabilities**

Using the same shop, find $P(X = 1)$, to three decimal places.

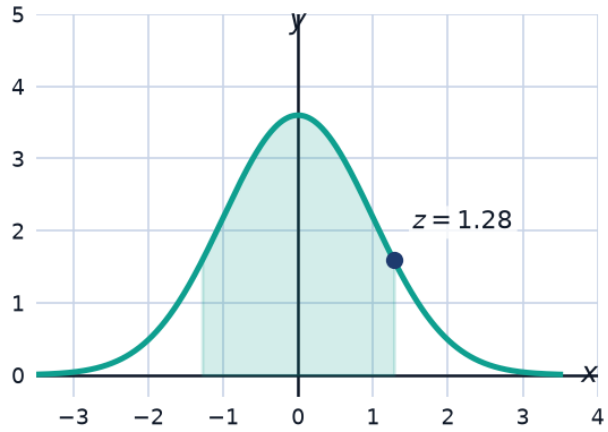
5 **[Foundational]** • **Target: Computing Poisson probabilities**

Using the same shop, find $P(X \leq 2)$.

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6 [Foundational] • Target: Mean and variance of a Poisson distribution

State the mean and variance of $X \sim \text{Po}(\lambda)$, and find the standard deviation when $\lambda = 9$.



7 [Foundational] • Target: Mean and variance of a Poisson distribution

A random variable $X \sim \text{Po}(7)$. State $E(X)$, $\text{Var}(X)$, and the standard deviation, to three decimal places.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Sums of independent Poisson variables

Two independent Poisson processes have rates $\lambda_1 = 4$ and $\lambda_2 = 6$ per hour. State the distribution of the total number of events, and find its mean and variance.

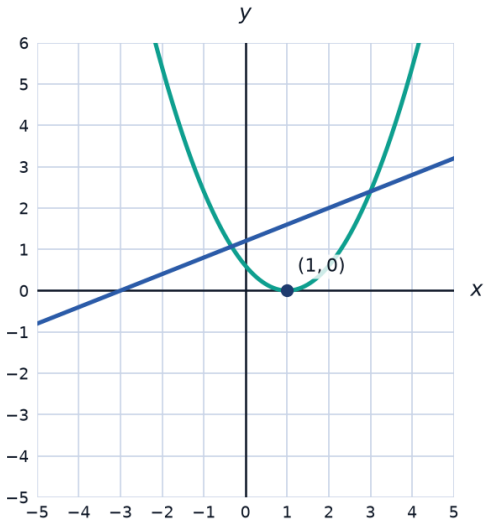
9 [Intermediate] • Target: Sums of independent Poisson variables

Three independent Poisson processes have rates $\lambda_1 = 2$, $\lambda_2 = 4$, $\lambda_3 = 5$ per hour. Find the mean and variance of the total number of events across all three processes in one hour.

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10 [Intermediate] • Target: Poisson as an approximation to the binomial

Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the value of λ to use.



11 [Intermediate] • Target: Poisson as an approximation to the binomial

A factory produces 2000 items, each independently defective with probability 0.0025. Using the Poisson approximation, find the probability that exactly 2 items are defective, to three decimal places.

12 [Intermediate] • Target: Poisson as an approximation to the binomial

A survey mails 5000 questionnaires, each independently containing a printing error with probability 0.0006. Using the Poisson approximation, find the probability that no questionnaires contain an error, to four decimal places.

13 [Intermediate] • Target: Using cumulative Poisson probabilities

A call center receives calls at $\lambda = 6$ per 10-minute interval. Find the probability of receiving more than 3 calls, given $P(X = 0) \approx 0.00248$, $P(X = 1) \approx 0.01487$, $P(X = 2) \approx 0.04462$, $P(X = 3) \approx 0.08924$.

14 [Intermediate] • Target: Using cumulative Poisson probabilities

A website receives visitors at $\lambda = 4$ per minute. Find $P(1 \le X \le 3)$, given $P(X = 0) \approx 0.0183$, $P(X = 1) \approx 0.0733$, $P(X = 2) \approx 0.1465$, $P(X = 3) \approx 0.1954$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

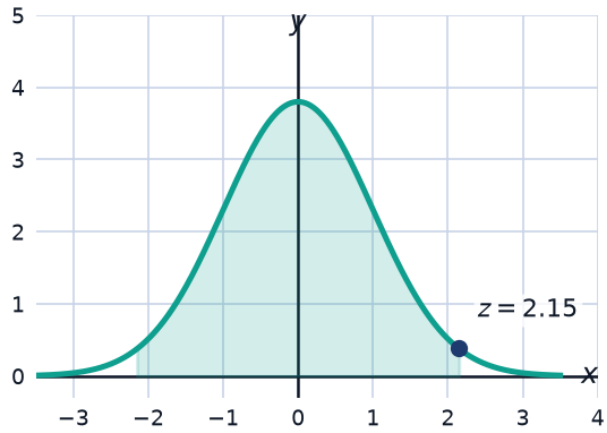
15 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 13.01 Continuous Random Variables

A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 0) = 0.0498$. Find λ, to the nearest whole number.

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16 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 14.01 Sampling & Estimation

A random variable $X \sim \text{Po}(\lambda)$ has standard deviation 2. Find λ, and then find $P(X = \lambda)$, to three decimal places.



17 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 15.01 Hypothesis Testing

A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 1) = 0.1494$, where λ is a whole number less than 5. Using $P(X = 1) = \lambda e^{-\lambda}$, determine λ by trial.

18 [Challenging] • Target: Poisson processes over different intervals and conditional probability • Cross-Lesson: 2.01 Exponentials & Logarithms

Machine faults occur at an average rate of $\lambda = 2$ per day. Find the probability that exactly 2 faults occur today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.1353$ and $P(X = 2) \approx 0.2707$.

19 [Challenging] • Target: Poisson processes over different intervals and conditional probability • Cross-Lesson: 3.01 Advanced Trigonometry

A radioactive source emits particles at a rate of $\lambda = 1.2$ per second, modeled by a Poisson distribution. Find $P(X = 2)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.

20 [Challenging] • Target: A well-designed word problem: quality control at a bottling plant • Cross-Lesson: 4.01 Differentiation

This question has four parts. Faulty caps occur at random in bottling cases with an average rate of $\lambda = 1.5$ per case. (a) Find $P(X = 0)$ and $P(X = 1)$ for a randomly selected case, to four decimal places. (b) Find the probability that a case has at least 2 faulty caps. (c) The plant inspects 40 cases per day. Using the fact that a sum of independent Poisson variables is itself Poisson, state the distribution of the total number of faulty caps found across all 40 cases in a day, and find its mean. (d) Since this total has a large λ, it can be approximated by $N(\lambda, \lambda)$. Using a continuity correction and $\Phi(1.36) \approx 0.9131$, estimate the probability that more than 70 faulty caps are found in a day.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Conditions for a Poisson distribution

A machine produces metal rods whose lengths are normally distributed with unknown mean μ cm and known standard deviation $\sigma = 0.8$ cm. A quality control manager tests the hypotheses $H_0 : \mu = 25.0 \quad \text{against} \quad H_1 : \mu < 25.0$ at the 5% level of significance. A random sample of 16 rods yields a sample mean length of $\bar{x} = 24.6$ cm. Given that the critical value for this test is $z_{\text{crit}} = -1.645$, which of the following gives the correct test statistic z and the corresponding conclusion?

a. $z = -2.0$; reject H_0 because $-2.0 < -1.645$

b. $z = -2.0$; do not reject H_0 because $-2.0 < -1.645$

c. $z = -0.5$; do not reject H_0 because $-0.5 > -1.645$

d. $z = -8.0$; reject H_0 because $-8.0 < -1.645$

22 [Foundational] • Target: Computing Poisson probabilities

$X \sim Po(3)$ and $Y \sim Po(3)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 9

b. 0

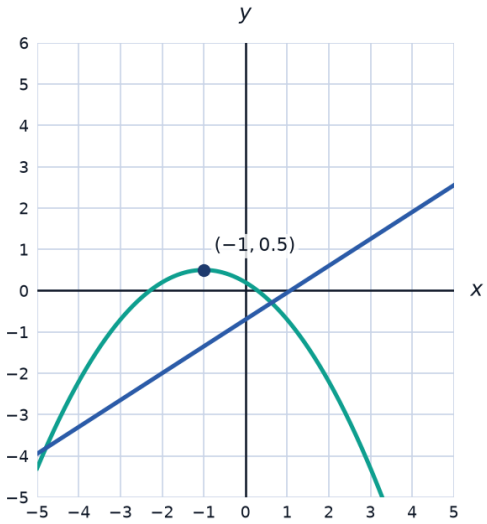
c. 6

d. 7

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23 [Foundational] • Target: Mean and variance of a Poisson distribution

A binomial random variable has $n = 200$, $p = \frac{1}{50}$, and is approximated by a Poisson distribution. Find the parameter λ of the approximating Poisson distribution.



a. 50

b. 4

c. 200

d. 5

24 [Foundational] • Target: Sums of independent Poisson variables

The random variable X has a Poisson distribution with mean $\lambda > 0$. Given that $3\,\mathrm{P}(X = 2) = \mathrm{P}(X = 4)$, find the value of λ.

a. 6

b. 36

c. 12

d. $\sqrt{6}$

25 [Foundational] • Target: Poisson as an approximation to the binomial

A factory produces light bulbs. The probability that a randomly selected light bulb is defective is p. The number of defective light bulbs in a random sample of n bulbs follows a binomial distribution $X \sim B(n, p)$. It is known that the mean number of defective bulbs in a sample of $n = 25$ is 2. \\ (i) Calculate the value of p. \\ (ii) For a sample of $n = 50$ bulbs, calculate the probability of finding exactly k defective bulbs, where k is the smallest integer such that $P(X \leq k) > 0.9$. \\ (iii) If a sample of $n = 50$ bulbs is taken, find the smallest value of n' such that the probability of observing at least one defective bulb is greater than 0.999 .

a. 0.04

b. 0.08

c. 0.12

d. 0.16

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26 [Foundational] • Target: Using cumulative Poisson probabilities

A random variable X follows a binomial distribution $B(n, p)$. The number of trials is $n = 10$ and the probability of success in a single trial is p. If the mean of X is 4, find the variance of X.

a. 2.4

b. 3.6

c. 6.4

d. 7.2

27 [Foundational] • Target: Finding an unknown parameter

Given $X \sim Po(7)$, find $P(X \geq 1)$ in exact form.

a. $1 - e^{7}$

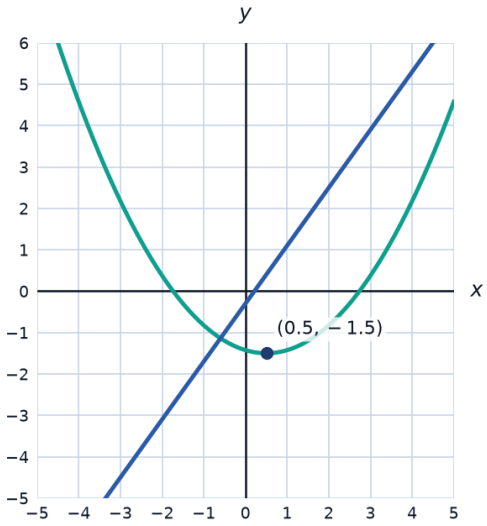
b. e^{-7}

c. $e^{7} - 1$

d. $1 - e^{-7}$

28 [Foundational] • Target: Poisson processes over different intervals and conditional probability

Given $X \sim Po(3)$, find an expression for $P(X = 5)$.



a. $\frac{3^6}{720}e^{-3}$

b. $\frac{3^4}{24}e^{-3}$

c. $\frac{3^5}{120}e^{3}$

d. $\frac{3^5}{120}e^{-3}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: quality control at a bottling plant

A factory tests 30 independent components, each with a fixed probability of being defective; X is the number of defective components. Is X binomially distributed?

a. Yes — fixed, independent trials with constant success probability

b. No — the number of trials is not fixed

c. No — trials are not independent / probability is not constant

d. Yes — but only if $n > 30$

30 [Intermediate] • Target: Conditions for a Poisson distribution

A random sample of 10 values of a variable x is taken. The results are summarised as follows: $\sum x = 50, \quad \sum x^2 = 322$ Calculate an unbiased estimate of the population variance.

a. 8

b. 7.2

c. 2.83

d. 32.2

31 [Intermediate] • Target: Computing Poisson probabilities

A random sample of $n = 50$ observations is taken from a population. The sample values are summarised as follows: $\sum x = 450, \quad \sum x^2 = 4295.$ Find an unbiased estimate of the population variance.

a. 5.0

b. 4.9

c. 2.24

d. 9.0

32 [Intermediate] • Target: Mean and variance of a Poisson distribution

Formulate the null and alternative hypotheses for the following scenario: "A factory claims the mean weight of a product is 250g. An inspector wants to test if it has changed (either direction)."

a. $H_0 : \mu = 500,\ H_1 : \mu < 500$

b. $H_0 : \mu = 3,\ H_1 : \mu > 3$

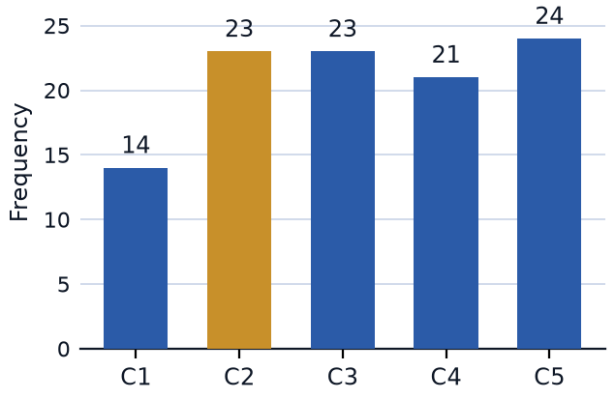
c. $H_0 : \mu = 10,\ H_1 : \mu < 10$

d. $H_0 : \mu = 250,\ H_1 : \mu e 250$

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33 [Intermediate] • Target: Sums of independent Poisson variables

Formulate the null and alternative hypotheses for the following scenario: "A manufacturer claims the mean lifetime of a battery is 500 hours. A researcher suspects it is less than this."



a. $H_0 : \mu = 500,\ H_1 : \mu < 500$

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$

c. $H_0 : \mu = 3,\ H_1 : \mu > 3$

d. $H_0 : \mu = 250,\ H_1 : \mu e 250$

34 [Intermediate] • Target: Poisson as an approximation to the binomial

Formulate the null and alternative hypotheses for the following scenario: "A school claims the mean test score is 70. A teacher believes it is actually higher."

a. $H_0 : \mu = 10,\ H_1 : \mu < 10$

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$

c. $H_0 : \mu = 500,\ H_1 : \mu < 500$

d. $H_0 : \mu = 3,\ H_1 : \mu > 3$

35 [Intermediate] • Target: Using cumulative Poisson probabilities

Using the trapezium rule with 3 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 1, y_1 = 1, y_2 = 2, y_3 = 10$.

a. $\frac{17}{2}$

b. 7

c. $\frac{103}{12}$

d. $\frac{19}{2}$

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36 [Intermediate] • Target: Finding an unknown parameter

Find $\int x(x^2+1)^2\,dx$ using substitution.

a. $\frac{x^2(x^2+1)^2}{2}+C$

b. $\frac{x^6}{6}+\frac{x^4}{2}+\frac{x^2}{2}+C$

c. $\frac{2(x^2+1)^3}{3}+C$

d. $\frac{(x^2+1)^3}{3}+C$

37 [Intermediate] • Target: Poisson processes over different intervals and conditional probability

A continuous random variable X has PDF $f(x)=\frac{2}{2^2}x$ for $0\le x\le 2$. Find the median of X in exact form.

a. $\frac{2\sqrt{2}}{2}$

b. $\frac{2\sqrt{2}}{4}$

c. $\frac{2}{2}$

d. $2\sqrt{2}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: quality control at a bottling plant • Cross-Lesson: 2.01 Exponentials & Logarithms

A sample of size $n=9$ is drawn from a population with mean $\mu=42$ and $\sigma=8$. Given $\bar{x}=\frac{118}{3}$, find $Z=\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$.

a. 0

b. 1

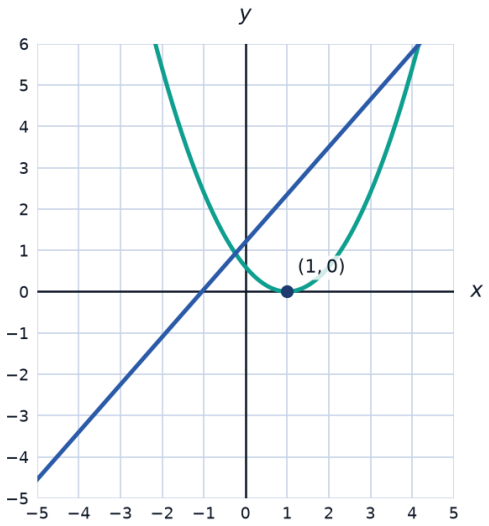
c. -1

d. -3

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39 [Challenging] • Target: Conditions for a Poisson distribution • Cross-Lesson: 3.01 Advanced Trigonometry

A continuous random variable X has PDF $f(x) = kx^1$ for $0 \le x \le 4$ (and 0 otherwise). Find k.



a. $\frac{5}{24}$

b. $\frac{1}{4}$

c. $\frac{1}{16}$

d. $\frac{1}{8}$

40 [Challenging] • Target: Computing Poisson probabilities • Cross-Lesson: 4.01 Differentiation

A continuous random variable X has PDF $f(x) = \frac{2}{4^2}x^1$ for $0 \le x \le 4$. Find $\text{Var}(X)$.

a. $\frac{17}{9}$

b. $\frac{8}{9}$

c. $\frac{8}{3}$

d. 8

41 [Challenging] • Target: Mean and variance of a Poisson distribution • Cross-Lesson: 5.01 Integration

A sample of size 6 has $\sum x = 282$ and $\sum x^2 = 13312$. Find the unbiased estimate of the population variance.

a. $\frac{6656}{3}$

b. $\frac{58}{5}$

c. $\frac{63}{5}$

d. $\frac{29}{3}$

42 [Challenging] • Target: Sums of independent Poisson variables • Cross-Lesson: 6.01 Sequences & Series

$X_1, \ldots, X_{14}$ are independent and identically distributed with $\text{Var}(X) = 3$. Find $\text{Var}(X_1 + \cdots + X_{14})$.

a. 42

b. 43

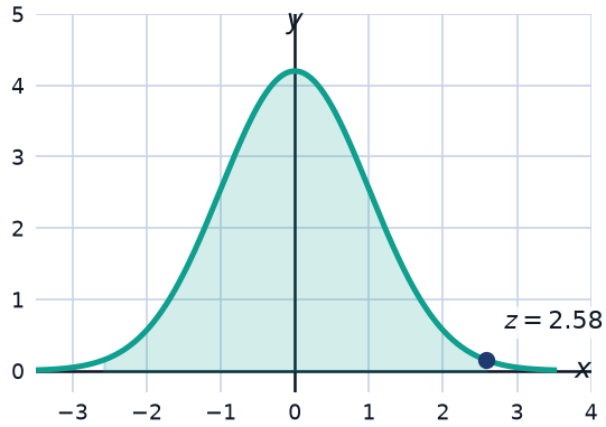
c. 3

d. 39

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43 [Challenging] • Target: Poisson as an approximation to the binomial • Cross-Lesson: 7.01 Vectors

A hypothesis test is carried out at the 10% significance level. Find the probability of a Type I error.



a. 90\%

b. 10\%

c. 50\%

d. 0\%

44 [Challenging] • Target: Using cumulative Poisson probabilities • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

A sample of size 6 gives values: 5, 12, 18, 11, 17, 18. Find the sample mean $\bar{x}$.

a. 18

b. $\frac{29}{2}$

c. $\frac{27}{2}$

d. $\frac{81}{7}$

45 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 9.01 Complex Numbers

A confidence interval for the mean uses $z = 1$, $\sigma = 8$, $n = 9$. Find the total width of the confidence interval, $2E$.

a. $\frac{19}{3}$

b. $\frac{16}{3}$

c. 8

d. $\frac{8}{3}$

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